Gate 7B - Final Job-Ready Fresh Retest

Protected fresh retest | Renewable-energy operations | Conditional RC1

Scenario

A renewable-energy operator receives solar-site work-order files, paginated inverter alerts and an operational asset database. Produce a trustworthy site-health platform using the same job-ready competencies but a new schema, new business keys, new quality rules and new recovery decisions.

Required work

- Read requirements and source contracts; declare grain, business keys and freshness expectations before coding.
- Ingest the supplied CSV + paginated API + operational database source and preserve raw evidence.
- Implement critical validation, quarantine and reconciliation. Bad data must not silently reach trusted output.
- Build a relational/analytics model with clear facts/dimensions or equivalent trusted grain.
- Use dbt-style dependency/tests or equivalent tested transformation evidence.
- Orchestrate the repeatable path with failure-safe dependencies, retries and bounded backfill/reprocess logic.
- Use PySpark/Delta only where the scale requirement justifies it; document partition/performance reasoning.
- Design the Microsoft Fabric/OneLake/Lakehouse deployment, access model, CI/CD and monitoring.
- Prove safe rerun/idempotency and explain one deliberate failure/recovery path.
- Commit/document the work as a reproducible engineering repository and complete a technical defense.
- Run the offline Gate checker and save the output. Live runtime evidence remains separate before FINAL.

Fresh constraints

- Do not copy Gate 7A field names or output tables.
- A repeated work-order drop must not double count.
- A sensor alert arrives late and one alert has an invalid severity.
- PII in technician contact data must be protected.
- A failed quality check must prevent trusted publish.

Protected rules

- Do not open Gate 7B Review before saving the retest.
- No real credentials.
- The retest is meant to prove transfer, not memory.